Brand Review Desk

Round 3 Final Project Submission & Product Verification

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Live Demo: https://brand-review-desk.vercel.app/ | **Source Code:** https://github.com/Tvaibhav06/code-axis | **Submission:** CodeAxis Round 3 (72h Final)

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1. EXECUTIVE SUMMARY & CORE MISSION

Brand Review Desk is an enterprise brand governance and campaign pre-flight review platform built in response to Inkloom AI's brand-governance brief. In modern marketing, multi-channel assets (ad copy, graphical banners, campaign briefs) are frequently deployed with outdated offer durations, off-brand claims, or conflicting terms. Brand Review Desk introduces an automated, explainable, and multi-modal verification loop that checks marketing materials before publication.

A non-negotiable architectural principle is that **AI assists, but humans decide**. The platform never automatically publishes campaigns; instead, it enforces an immutable approval transaction binding a specific campaign revision, published rulebook version, and check run. In our founding SampleCo scenario, draft copy claims a "30 days free trial" and "guaranteed 50% savings" while the approved brand rulebook specifies 14 days and prohibits unverified guarantees. The system flags the conflicts, quotes offending spans, references source facts, checks visual banner text, parses brief documents, and produces an exportable, self-contained HTML audit record.

2. COMPLETE LIST OF FEATURES BUILT

Functional Area	Capability & Implementation Details	Status
Multi-Role Workspace	Pre-provisioned isolated workspaces with authenticated Creator, Manager, and Reviewer identities enforced by Supabase Auth and Row Level Security.	**Complete**
Versioned Rulebooks	Managers publish immutable source snapshots (facts, terminology, aliases, logo asset, and palette) with cryptographic version increments.	**Complete**
Immutable Revisions	Creators edit campaign copy, title, purpose, and audience. Each save creates a numbered revision with full historical tracking.	**Complete**
Image OCR (Round 3)	Server-side Tesseract.js OCR extracts text inside PNG/JPG images with human correction/confirmation preview and cross-modal claim verification.	**Complete**
PDF Review (Round 3)	Private PDF brief ingestion (up to 5 pages, 2 MB, 1,500 words) with page-mapped text extraction, preview, and compliance checks.	**Complete**
Guideline Import (Round 3)	Managers extract rule/fact candidates from PDF briefs with verbatim span validation and publish them directly to versioned rulebooks.	**Complete**
Explainable Checks	Dual check engine: high-speed deterministic TypeScript rules + optional Gemini 2.5 Flash REST integration with strict Zod structured outputs.	**Complete**
Human Review Desk	Reviewers resolve/dismiss findings, log clarification requests, verify visual contrast/spacing checklists, and record reasoned decisions.	**Complete**
Approval Locking	SQL row-level locking ensures approvals bind strictly to the reviewed content revision; auto-invalidates when copy or sources change.	**Complete**
Audit Handoff	Downloadable, self-contained HTML audit package with cryptographic signatures, timestamped reviews, findings, and rule citations.	**Complete**

3. DELIVERY OF THE THREE ROUND 3 COMMITMENTS

In Round 2, DIZZY DEV committed to delivering three substantial capabilities during the final 72-hour window. All three have been fully implemented, verified, and hardened:

Improvement 1: Review Text Inside Images (OCR)

- **Engine & Scope:** Integrated server-side OCR via Tesseract.js supporting still PNG and JPG images up to 5 MB and 4,096 pixels per side.
- **Human Confirmation:** The creator clicks "Read text from this image," reviews the recognized text in an editable preview, and confirms it before claims enter the review pipeline.
- **Cross-Modal Verification:** The engine flags multi-modal discrepancies where text copy and image text contradict each other (e.g., copy claims 30 days while image banner promises 14 days).
- **Acceptance Target:** Achieved **10/10 passing fixtures** on our benchmark test suite (exceeding the 8/10 target), with unreadable text properly routed to mandatory manual review.

Improvement 2: Review Short Text PDFs

- **Native Stream Parsing:** Integrated server-side PDF text extraction via `pdfjs-dist` and `pdf-parse`, extracting per-page text maps (Page 1, Page 2).
- **Boundary Protection:** Strictly enforces limits (up to 5 pages, 2 MB, 1,500 extracted words). Rejects scanned non-text PDFs, encrypted files, and oversized documents with actionable error messages.
- **Explainable Page References:** Violations in supporting documents explicitly cite the exact page number (e.g., "Violates Rule on Page 2").
- **Full Workflow Integration:** Extracted document text is saved in campaign versions and checked by both exact rules and AI checks alongside copy text.

Improvement 3: Import Guidelines with Manager Confirmation

- **Candidate Extraction:** Reuses document parsing to propose up to 10 structured rule candidates (trial days, pricing, terminology, tone) from uploaded brand PDFs.
- **Anti-Hallucination Guard:** Every candidate is programmatically verified as a verbatim substring of the PDF text, preventing injected or hallucinated rules.
- **Manager Verification:** Managers can edit candidate titles/values, assign owners and validity dates, discard non-rules, and explicitly publish.
- **Approval Invalidation:** Publishing a newly imported rulebook increments the database version, immediately invalidating prior campaign approvals.

4. RELIABILITY, RESILIENCE & USABILITY HARDENING

A core evaluation criterion for Round 3 is product reliability without developer intervention. We resolved critical real-world failure modes:

- **Atomic Server-Side Uploads:** Re-architected image and PDF uploads through `/api/assets/upload` and `/api/assets/upload-pdf`, completely eliminating browser-to-storage CORS preflight failures and ambient cookie serialization errors.
- **Trans-Continental Network Resilience:** Diagnosed remote Supabase Auth latency (~11s) and expanded network timeouts from 10s to 30s across `supabase.ts`, `proxy.ts`, and `demo.ts`, eliminating premature 503 aborts.
- **Concurrency & Double-Click Locks:** Added immediate synchronous busy locks on demo initialization, role switching, and version saves to eliminate race conditions.
- **Demo Pool Auto-Recycling:** Added the `recycle_expired_demo_slots` migration function to reclaim expired demo workspaces, ensuring visitors never hit exhausted pool capacity.
- **Deterministic Fail-Open Architecture:** If external AI calls fail or exhaust provider quotas, the application fails open cleanly to deterministic TypeScript rule checking and manual checklists. The user journey is never blocked.

5. TECHNICAL ARCHITECTURE & TECH STACK

Layer	Technology Used	Design Choice & System Responsibility
Application & UI	Next.js 16.4.2 (App Router), React 19, TypeScript	Server-rendered shell with dynamic client hydration; handles routing, sessions, and fast local navigation.
Database & Identity	Supabase Postgres, Row Level Security	Strict workspace isolation enforced by Postgres RLS policies; transactional row-locking for approvals.
Asset Storage	Private Storage (`brand-assets`, `brand-documents`)	Zero public access; files are uploaded server-side and served via short-lived signed URLs.

Layer	Technology Used	Design Choice & System Responsibility
OCR & Processing	Tesseract.js, Sharp	Sharp enforces 4,096px / 16MP limits and decoding safety; Tesseract extracts English image text.
Document Ingestion	pdfjs-dist, pdf-parse	Server-only dynamic imports extract per-page text streams without client bundle bloat.
AI Compliance	Gemini 2.5 Flash, Zod	Server-side REST requests validated with strict Zod schemas requiring quoted spans and source IDs.
Hosting & Edge	Vercel / Render.com (Node 22)	Global CDN, edge middleware, and HTTPS termination; fully containerized Node 22 execution.

6. TESTING DETAILS & VERIFIED EVIDENCE

- **67/67 Unit & Rule Tests Passed:** 100% pass rate across 6 test suites: `ocr-fixtures.test.ts` (10/10 fixtures passing), `pdf-extract.test.ts` (11/11 tests passing), `conflicts.test.ts` (2/2), `checks.test.ts` (9/9), `content-inputs.test.ts` (21/21), and `images-handoff.test.ts` (19/19).
- **Permission & Security Verification:** The real-account Supabase permission suite passed with zero failures, proving cross-workspace isolation, immutable history preservation, and storage security.
- **Type & Build Verification:** Clean compilation under `npm run typecheck` and optimized production build under `npm run build`.

7. PRODUCT ROADMAP & FUTURE SCOPE

The architectural foundation built in Round 3 positions Brand Review Desk for enterprise expansion:

Milestone 1: Multi-Tenant Enterprise Workspaces & Custom Brand Portals (Q1 2027)

- **Self-Service Company Instances:** Allow organizations to register their own dedicated company workspaces with custom subdomains (e.g., `review.brand.com`) and enterprise SAML/SSO integration.
- **Multi-Campaign Architecture:** Support multiple concurrent campaign tracks, product lines, and sub-brands under unified organizational guidelines.
- **Granular Role-Based Access:** Introduce specialized roles for External Creative Agencies, Legal Directors, and Compliance Auditors.

Milestone 2: Enterprise AI Headroom & Multi-Model Resilience (Q2 2027)

- **Dedicated Enterprise AI Key & Higher Rate Limits:** Provision dedicated enterprise-tier API keys (Vertex AI / Gemini Paid Tier) with guaranteed throughput SLAs and high concurrency headroom.
- **Multi-Model Fallback Mesh:** Automatic failover across independent frontier models (Gemini 2.5 Flash → Claude 3.5 Sonnet → GPT-4o) when provider latency exceeds 5 seconds.
- **Adaptive Brand Learning:** Fine-tune embeddings on historical reviewer feedback to catch subtle nuances in brand tone and style.

Milestone 3: Expanded Ingestion & Direct Marketing Integrations (Q3 2027)

- **DOCX & Creative Ingestion:** Extend file ingestion to parse Microsoft Word (`.docx`) briefs and Figma design frames.
- **Direct Publishing Webhooks:** Export cryptographically approved review bundles directly to CMS platforms (Contentful, WordPress) and ad networks (Google Ads, Meta Ads Manager) upon final approval.
- **Multilingual Compliance:** Expand OCR and rule checks to multilingual global campaigns (Spanish, German, French, Hindi, Japanese).

8. JUDGE WALKTHROUGH & QUICK VERIFICATION GUIDE

1. **Access Demo:** Open https://brand-review-desk.vercel.app/ and click **"Start demo"** (instantly leases an isolated workspace as Creator).
2. **Test Image OCR:** Upload `docs/manual-fixtures/sampleco-campaign.png`, click **"Read text from this image"**, verify extracted text ("*Try SampleCo free for 14 days.*"), and confirm.

3. **Test PDF Extraction:** Under Campaign Brief / PDF, upload `docs/manual-fixtures/sample-guidelines.pdf`. Review the 2-page text preview. Save version and run review checks.
4. **Test Guideline Import:** Switch to **Manager**. In Sources, click **"Import guidelines from PDF"** with `sample-guidelines.pdf`. Inspect extracted rule cards and click **"Publish source version"**.
5. **Test Approval & Export:** Switch to **Reviewer**. Notice checks re-evaluated against the new rulebook. Complete manual checkboxes, click **"Approve version"**, and download the signed HTML audit package.

9. AUTHORSHIP DISCLOSURE & ATTRIBUTION

DIZZY DEV is a solo participant project by **Vaibhav Tandon** (B.Tech, The LNM Institute of Information Technology, Jaipur). In accordance with event guidelines, extensive AI assistance from OpenAI Codex and Google DeepMind's Antigravity coding agents is openly disclosed for code generation, migration scripts, and documentation support. All architectural decisions, security boundaries, debugging, and verification were directed and validated by the author.